TURKIC INTERNATIONAL MATHEMATICAL OLYMPIAD

TIMO

GRADE 4

Mathematical Horizons: Uniting Turkic Minds, Expanding Global Genius.

SCHOLARSHIP FROM TURKISH UNIVERSITIES

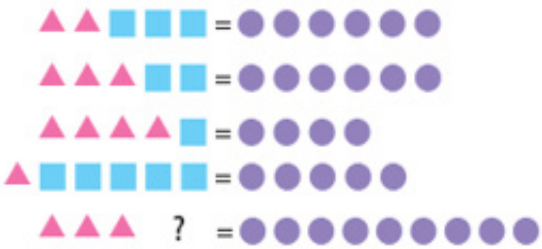
MEDIPOL · BAHÇEŞEHİR · İSTANBUL · İSTANBUL GALATA UNIVERSİTESİ

1. $\square \times (\square \times \square - \square) = 28$

By placing 4 of the digits 1, 2, 3, 4 and 5 in the boxes above, the given equality will be ensured.
If the digits in the boxes above different from each other what is the smallest digit which is no used?

A) 1 B) 2 C) 3 D) 4 E) 5

2. By applying a special rule with the given symbols " ▲ " and " ■ ", the symbols " ● " are obtained.



If Since the same rule is valid in the fifth equation, what is the number of symbols " ■ " that should be written instead of the question mark?

A) 2 B) 3 C) 4 D) 5 E) 6

3. If a number doesn't change when it is written in the reverse order than this number is called as a Palindrome number.
For example 22, 636, 4004 are two, three and four digit Palindrome numbers respectively.
According to the information given above how many 3 digit Palindrome number whose sum of digits is 7 are there?

A) 3 B) 4 C) 5 D) 9 E) 16

4. 224444666666888888888101010 …

Starting from the smallest one, by writing positive even numbers side by side, a large number having more than 255 digit was formed as shown above.

What is the 255th digit of this number?

A) 2 B) 4 C) 5 D) 6 E) 8

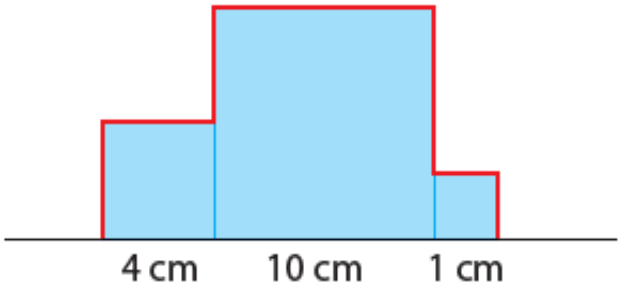
5.

The image above consists of 6 filled circles.

By moving (i.e. moving) at least how many circles we can arrange the whole shape upside down?

A) 1 B) 2 C) 3 D) 4 E) 5

6.



The above figure is obtained by placing 3 different squares side by side on a flat surface.

If the lengths of the sides of the squares are 4 cm, 10 cm, 1 cm respectively, what is the sum of the lengths shown in red?

A) 27 B) 30 C) 35 D) 36 E) 38

7. Ayşe writes a 3-digit natural number in her notebook and gives the following informations about this number to her friends.

- The number is an odd number
- The sum of the digits of the number is 7

In order for her friends to write the same number with Ayşe definitely, at least howmany number should they write?

A) 6 B) 8 C) 9 D) 10 E) 12

8. When a three digit natural number is multiplied by 3 and divided by 3 if the result is also a three digit natural number then we call this number as TMY number.
Howmany TMY numbers are there?

A) 8 B) 10 C) 12 D) 14 E) 16

9. Among three friends, Ali, Ahmet and Ayşe, Ali can read 50 pages of books in 2 days, Ahmet 80 pages in 3 days and Ayşe 100 pages in 5 days.

According to this, how many pages can these three friends read in 12 days?

A) 500 B) 580 C) 640 D) 720 E) 860

10. AB is a two digit and CCC is a three digit natural number. Given

$$\begin{array}{r} AB \\ CCC \\ +\ CCC \\ \hline 2024 \end{array}$$

What is the value of the sum A+B?

A) 7 B) 8 C) 9 D) 10 E) 11

11. The number of apples in the following 3 packs is written on them.

Emine wants to equalise the number of apples in the three packages by putting apples in each of them.
If she wants to make the total number of apples in the three packs more than 50 apples, minimum how many apples should she put in total?

A) 32 B) 35 C) 37 D) 38 E) 41

12.Four friends named Azra, Beril, Ceylan and Didem had a 10-day holiday together. Throughout the holiday, every morning one of them paid for the breakfast they had together.

Azra paid for 3 days, Beril and Ceylan did not pay for 8 days 6 days respectively. According to this, how many days did Didem pay for breakfast?

A) 0 B) 1 C) 2 D) 3 E) 4

13. Using a 76 mm long wire, Ahmet makes two squares whose side lengths are integers in mm. Then he realises that he can completely fill the big square with smaller squares equal to the small square he made.

According to this, at most how many identical smaller squares can Ahmet fit into the large square he made?

A) 225 B) 256 C) 289 D) 324 E) 358

14. In a chess tournament with 20 participants, each player was paired with one other player and played one round of matches. If the match ended in a draw, each player was awarded one gold coin, and if one of the players was victorious, the victor was awarded 3 gold coins. The defeated player did not receive any prize.

If the total number of coins distributed at the end of the tournament was 24, how many matches ended in a draw?

A) 3 B) 4 C) 5 D) 6 E) 7

15. A magic square whose sum of three numbers written on the same row, column and diagonal is equal to each other is given below

	5	
4	9	

Except three of the numbers in this square, all the others are deleted.
According to this, what is the sum of the deleted numbers?

A) 25 B) 27 C) 36 D) 38 E) 40

16. For two digits natural numbers AB and BA the equation AB+B=BA satisfies. According to this, what is the difference between BA and AB ?

A) 9 B) 18 C) 27 D) 36 E) 45

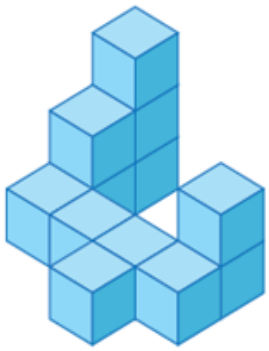
17. Aysel keeps a two-digit natural number in her mind and she gives the following information about this number.

- The tens digit of the number is divisible by the unit digit of the number
- The product of the number's digit is 27

According to this, what is the sum of the digits of the number she keeps in her mind?

A) 12 B) 11 C) 10 D) 9 E) 8

18. An object made of unit cubes is given below.



In order to turn this object into a filled cube, at least how many more unit cubes are needed?

A) 42 B) 44 C) 48 D) 50 E) 52

19. In each row and each column of the 3×3 table given below, there will be the numbers 1, 2 and 3. Some numbers are written below.

		2
1		

According to this what is the sum of the numbers that will be writen on the colored cells?

A) 4 B) 5 C) 6 D) 8 E) 9

20.

7 numbers written side by side on a piece of paper are given above. This paper is cut in two places corresponding to the gaps between the numbers and 3 pieces of paper are obtained. Then the numbers on these three pieces of paper are added.
For example if the paper is cut as shown below

the numbers obtained are 12, 503 and 47 and the sum of these numbers is 562.
What is the smallest value of the sum of the three numbers obtained by cutting this paper in two places?

A) 562 B) 357 C) 175 D) 166 E) 143